LAB 04 - PROMPTS

Hypothesis Stage: from a Written Card to an Automated Statistical Test

Open your AI coding assistant (for example Claude Code or GitHub Copilot) in the labs_fullcode folder. Work down the steps: copy each prompt into the assistant, review the code it writes, then run the command. The complete, working code is already in Lab04-trading-hypothesis/solution/ if you want to compare. Never paste API keys into an AI chat.

Step 1: Read the hypothesis card (A1, K1)

Do this (no prompt needed)
Open pipeline/hypotheses/spy_dip_after_streak.yaml. Identify H1, H0, the signal (3 consecutive down closes, known at the close), the outcome (next-day return), the test and the decision rule (p < 0.05 AND an edge of at least 3 bps, the estimated round-trip cost).

Check: You can say why the outcome must be a FUTURE return and why the decision rule includes a cost hurdle.

Step 2: Write a card for your own idea (A1, K1)

Prompt - copy into your AI coding assistant
Help me write a hypothesis card in YAML for this trading idea: <describe your idea in one sentence>. Use exactly the fields of labs_fullcode/pipeline/hypotheses/spy_dip_after_streak.yaml (id, name, instrument, timeframe, sample, rationale, h1, h0, signal, outcome, test, decision_rule, robustness, biases_to_watch, big_data_use_case). The signal type must be one of losing_streak (n), rsi_below (window, level) or close_below_sma (window) and must be known at the close of day t; the outcome must be a forward return over horizon_days; H0 must be 'no difference'. Do not analyse any data.

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Check: my_card.yaml has an economic rationale, H1, H0, a signal known at the close and a decision rule.

Step 3: Build the test engine (A1)

Prompt - copy into your AI coding assistant
Create labs_fullcode/pipeline/hypothesis.py. losing_streak(close) counts consecutive down closes ending at each day. condition(bars, signal) builds the boolean signal for the types losing_streak, rsi_below (RSI from common.indicators) and close_below_sma, using data up to the close of day t only. test_card(bars, card, signal=None): outcome fwd = close.shift(-horizon) / close - 1 (the outcome, never a feature); compare signal days with all other days: counts, means, hit rates, scipy.stats.ttest_ind(equal_var=False), a bootstrap 95% CI of the signal-day mean with numpy default_rng(card seed) and card bootstrap draws, edge_bps = (mean signal - mean other) x 10,000, supported = p < alpha AND t > 0 AND edge_bps >= min_edge_bps. Load bars with pipeline.ingest.load_bars. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: test_card returns a dict with signal_days, means, ci95, welch_t, p_value, edge_bps and supported.

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Step 4: Add the robustness grid, chart and result file (A1, A6)

Prompt - copy into your AI coding assistant
In labs_fullcode/pipeline/hypothesis.py add run(card_path=None) for `python -m pipeline hypothesis [--card PATH]` (default pipeline/hypotheses/spy_dip_after_streak.yaml): print H1, H0, sample, signal days, means, hit rates, CI, Welch t-test, edge and a DECISION line ('SUPPORTED - reject H0; promote the idea to the signal and backtest stages' or 'NOT SUPPORTED - do not reject H0'); a robustness grid over card.robustness.streaks for the instrument and each robustness symbol with a Bonferroni threshold alpha / number of variants; a table of forward return by streak length (0..4, 5+) with standard errors; chart Lab04_hypothesis.png (bars with 95% error bars, unconditional mean line); and write the card plus results to _outputs/Lab04_hypothesis_result.yaml. Rules: run from the labs folder; reuse labs_fullcode/common (config, data, broker, indicators, metrics, backtest, plotting) and earlier pipeline stages instead of rewriting them; do not modify common/ or data/; read every setting from pipeline/config.yaml via pipeline.config.load_config(); never put API keys or secrets in code; paper trading only; keep a main(argv) function so `python -m pipeline <command>` works; add short comments a finance professional can follow.

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Check: python -m pipeline hypothesis --help shows --card.

Step 5: Run the test on the desk's card (A1)

Do this (no prompt needed)
Run: python -m pipeline hypothesis (same as: python3 Lab04-trading-hypothesis/hypothesis_stage.py).

Check: SPY 2018-01-02 to 2025-12-30 (2,010 days): 178 signal days (8.9%); mean next-day return +0.2828% vs +0.0384% on other days (unconditional +0.0601%); hit rate 56.7% vs 55.4%; bootstrap CI [+0.0531%, +0.5136%]; Welch t = 2.03, p = 0.044; edge +24.4 bps; 'DECISION: SUPPORTED'.

Step 6: Read the robustness grid (A1, K1)

Do this (no prompt needed)
Read the robustness table and the Bonferroni line. Which variants would survive if the desk counted every test it ran?

Check: 3 of 6 variants supported: SPY n=3 (p 0.044), QQQ n=2 (p 0.031) and QQQ n=3 (p 0.006); SPY n=2 (p 0.296), SPY n=4 (p 0.053) and QQQ n=4 are not. Bonferroni threshold p < 0.0083 - only QQQ n=3 passes it.

Step 7: Study the shape of the effect (A1)

Do this (no prompt needed)
Read the table 'Forward return by losing-streak length' and open _outputs/Lab04_hypothesis.png.

Check: Streaks of 4 (49 days, +0.439%) and 5+ (27 days, +0.533%) look strongest but have the widest error bars - small samples.

Step 8: Test your own card and its big-data use case (A1, A6)

Prompt - copy into your AI coding assistant
Review my hypothesis result in labs_fullcode/_outputs/Lab04_hypothesis_result.yaml. In plain English, say whether the evidence supports H1, how many variants I tested, whether the result survives a Bonferroni correction, and which bigger or alternative data set (volume, velocity or variety) would test the idea better. Do not change any code.

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Check: Your card prints a DECISION line and _outputs/Lab04_hypothesis_result.yaml is rewritten for your card.

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Run it

Step	Command / file
Run Lab 04	python3 Lab04-trading-hypothesis/hypothesis_stage.py (same as: python -m pipeline hypothesis)
The complete code	pipeline/hypothesis.py (reference copy: Lab04-trading-hypothesis/solution/)
Outputs	_outputs/ (charts, CSV files and reports written by the stage)

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